# Department of Computer & Software Engineering

# CE110L - Electronics Workbench Lab

## Lab Project

ARDUINO LINE FOLLOWING AND OBSTACLE AVOIDING ROBOT


## Submitted by:

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**Subject: Collaborative Project Report - Dual Mode Line Following and Obstacle Avoiding Robot**

Dear SIR JUNAID ASHRAF,

We, the undersigned students, are submitting this collaborative report for our project on building a Dual Mode Line Following and Obstacle Avoiding Robot. This project is a significant part of our course and carries a weightage of 15% in our final grade. We appreciate the opportunity to work on this project collaboratively, and we believe that the diverse skills and contributions from each team member have led to a successful outcome.

## Title:

Build a Dual Mode Line Following and Obstacle Avoiding Robot Using the Robot Car Kit.

## Mode Selection:

The selection of the mode is controlled by a push button. When the push button is pressed, the working state of the robot will change from line following to obstacle avoiding, and vice versa.

# Project Components

## 1. Physical Structure and Electrical Wiring

*Description:*

The physical structure and electrical wiring of the robot are crucial for the project's success. It includes the proper assembly of the robot car kit components and the correct wiring of motors, sensors, push button, LED, and other electronic components.

## 2. Programming

*Description:*

The provided code controls the behavior of the robot. It incorporates the logic for both line following and obstacle avoiding modes, as well as the switch between these modes based on the push button input.

## 3. Performance of the Robot

*Description:*

The performance of the robot is assessed based on its ability to follow lines and avoid obstacles. The robot should demonstrate smooth transitions between modes and effective obstacle avoidance.

1. **Obstacle Avoidance:**
    - **Description:** When the robot is in obstacle avoidance mode (triggered by releasing the push button), it stops its regular line-following behavior and focuses on avoiding obstacles in its path using an ultrasonic sensor.
    - **Logic:**
        - The robot constantly measures the distance to the nearest obstacle using the ultrasonic sensor.
        - If the measured distance is less than 30 units (adjustable threshold), it signifies the presence of an obstacle.
        - In response to detecting an obstacle, the robot initiates a turn to the right, allowing it to avoid the obstacle and find an alternative path.
    - **Expected Behavior:** The robot should effectively detect and navigate around obstacles, ensuring a safe and obstacle-free movement.
2. **Line Following:**
    - **Description:** When the push button is pressed, the robot switches to line-following mode, where it follows a predefined path marked by a contrasting line on the ground.
    - **Logic:**
        - The robot continuously monitors the signals from two line-following sensors, typically positioned on the left and right sides of the robot.
        - Based on the sensor inputs, the robot adjusts its motor speeds to stay on the line.
        - If both sensors detect the line, the robot moves forward. If only the right or left sensor detects the line, it turns in the corresponding direction.
        - If neither sensor detects the line, the robot stops, indicating it has reached the end of the line or lost the track.
    - **Expected Behavior:** The robot should smoothly follow the marked path, making necessary adjustments to stay aligned with the line.

In summary, the obstacle avoidance mode prioritizes detecting and avoiding obstacles to ensure the robot's safety, while the line-following mode guides the robot along a predefined path marked by a contrasting line on the surface. The ability to seamlessly switch between these modes allows the robot to adapt to different scenarios and navigate effectively in various environments

## Code Review

We programmed the code in Arduino, incorporating logic for both line-following and obstacle-avoiding modes. The code allows for seamless transitions between these modes based on the push button input.

```
const int in1 = 1;

const int in2 = 2;

const int in3 = 3;

const int in4 = 4;

const int ENA = 5;

const int ENB = 6;

const int right = A0;

const int left = A1;

const int button = 7;

const int led = 13;

const int trig = 9;

const int echo = 8;

long distance;


void setup() {

  pinMode(right, INPUT);

  pinMode(left, INPUT);

  pinMode(button, INPUT_PULLUP);

  pinMode(led, OUTPUT);

  pinMode(in1, OUTPUT);

  pinMode(in2, OUTPUT);

  pinMode(in3, OUTPUT);
```

```
  pinMode(in4, OUTPUT);
  pinMode(ENA, OUTPUT);
  pinMode(ENB, OUTPUT);
  pinMode(echo, INPUT);
  pinMode(trig, OUTPUT);
  Serial.begin(115200);
}

void loop() {
  int state = digitalRead(button);

  if (state == 0) {
    digitalWrite(led, HIGH);
    handleButtonPress();
  } else {
    digitalWrite(led, LOW);
    handleObstacleAvoidance();
  }
}

void handleButtonPress() {
  if (digitalRead(right) && digitalRead(left)) {
    stopMotors();
    delay(70);
  } else if (digitalRead(right) && !digitalRead(left)) {
    stopMotors();
    turn_left(80);
```

```
        delay(70);
    } else if (digitalRead(left) && !digitalRead(right)) {
        stopMotors();
        turn_right(80);
        delay(70);
    } else if (!digitalRead(right) && !digitalRead(left)) {
        stopMotors();
        run_forward(80);
        delay(70);
    }
}

void handleObstacleAvoidance() {
    stopMotors();
    distance = measure();
    Serial.println(distance);

    if (distance < 30) {
        delay(1000);
        turn_right(100);
        delay(500);
    } else {
        run_forward(80);
        delay(70);
    }
}
```

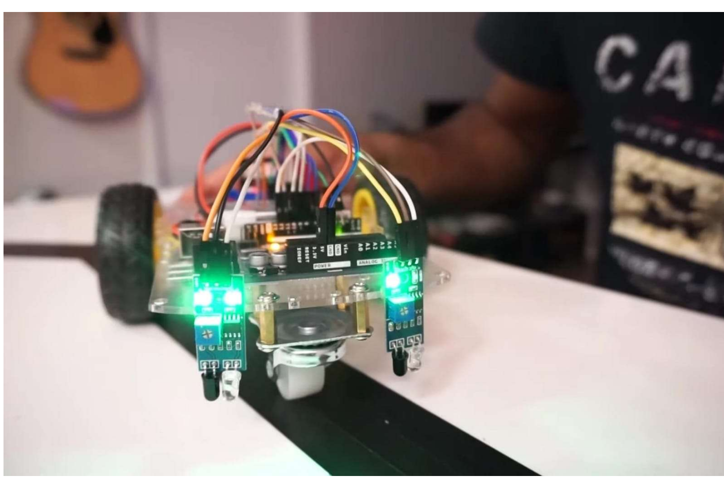
```
void stopMotors() {
  digitalWrite(in1, LOW);
  digitalWrite(in2, LOW);
  digitalWrite(in3, LOW);
  digitalWrite(in4, LOW);
  digitalWrite(ENA, LOW);
  digitalWrite(ENB, LOW);
}

long measure() {
  digitalWrite(trig, LOW);
  delay(10);
  digitalWrite(trig, HIGH);
  delay(10);
  digitalWrite(trig, LOW);
  long duration = pulseIn(echo, HIGH);
  return duration / 29 / 2;
}

void run_forward(int fwd_speed) {
    stopMotors();
    digitalWrite(in2, HIGH);
    digitalWrite(in4, HIGH);
    analogWrite(ENA, fwd_speed);
    analogWrite(ENB, fwd_speed * 2);
}

void turn_left(int left_speed) {
```

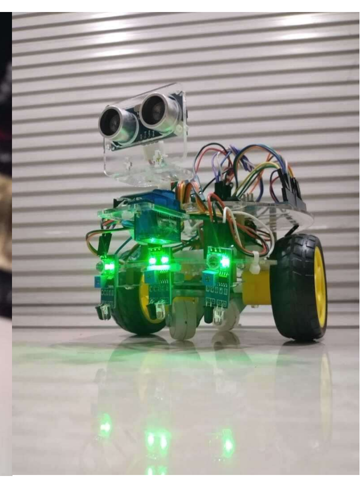
```
    stopMotors();
    digitalWrite(in2, HIGH);
    digitalWrite(in4, HIGH);
    analogWrite(ENA, left_speed / 4);
    analogWrite(ENB, 0);
}

void turn_right(int right_speed) {
    stopMotors();
    digitalWrite(in2, HIGH);
    digitalWrite(in4, HIGH);
    analogWrite(ENA, 0);
    analogWrite(ENB, right_speed * 2);
}
```

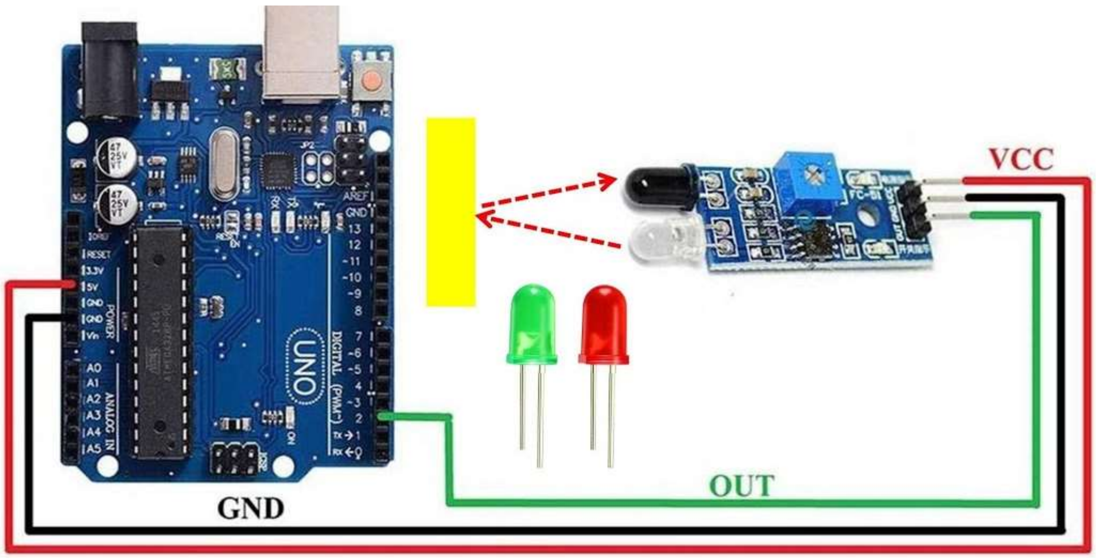
### *Key Code Components:*

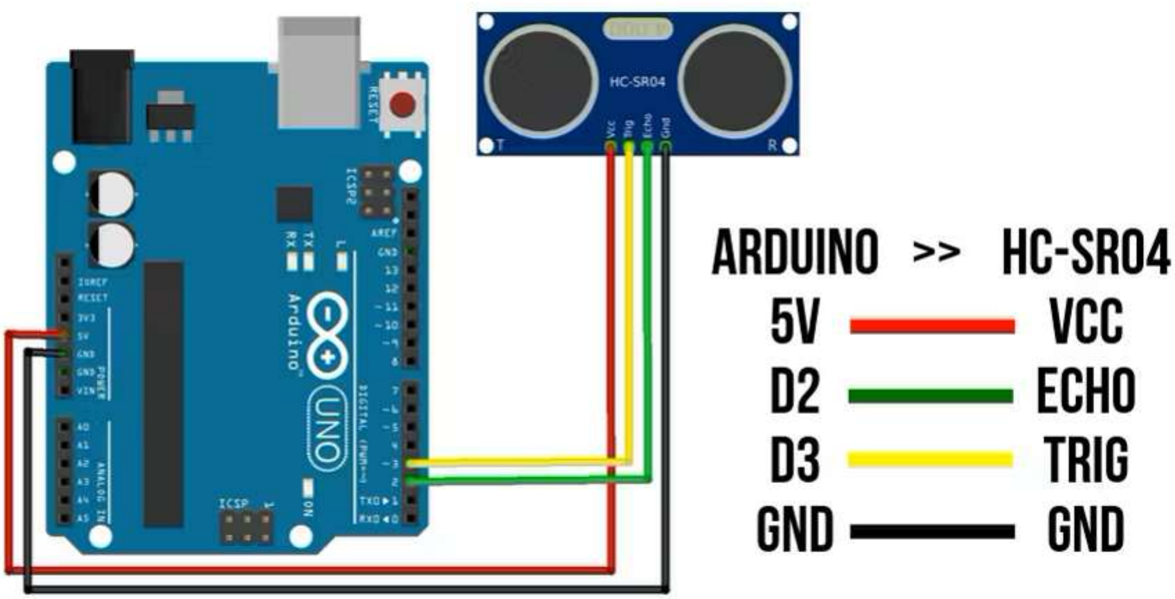
- Motor control logic for forward, left turn, and right turn.
- Mode switching based on push button state.
- Obstacle avoidance logic using ultrasonic distance measurement.

## Additional Documents






VCC
OUT
GND


HC-SR04
ARDUINO >> HC-SR04
5V VCC
D2 ECHO
D3 TRIG
GND GND

## Group Members:

1. **Mechanical Assembly and Wiring Specialist (BSCE23041):**
   - Assemble the physical structure of the robot car kit.
   - Ensure all components are securely in place and correctly connected.
   - Organize and manage the electrical wiring to guarantee proper connectivity.
   - Document the assembly process with clear photos for inclusion in the project report.
   - Focus on the robot's ability to follow lines and avoid obstacles in both modes.
2. **Programming and Code Implementation (BSCE23048):**
   - Implement the code in the Arduino IDE.
   - Incorporate the logic for both line following and obstacle avoiding modes.
   - Ensure the code allows for seamless transitions between modes based on the push button input.
   - Test the code on the robot to verify its functionality.
3. **Performance Testing and Documentation (BSCE23017):**
   - Conduct extensive testing of the robot's performance.
   - Gathered components used in the project.
   - Document the testing process and outcomes.
   - Created logic behind the program for the project task.
   - Document the assembly process for inclusion in the project report.

## Conclusion

In conclusion, our collaborative effort in designing the Dual Mode Line Following and Obstacle Avoiding Robot has yielded a successful and well-rounded project. Through meticulous physical assembly, adept programming, and comprehensive performance testing, each team member played a crucial role in achieving a robot that seamlessly transitions between line-following and obstacle-avoidance modes. The documented process, supported by clear visuals, underscores our commitment to precision and innovation. As a team, we have not only met the academic requirements but also cultivated valuable skills in communication, teamwork, and problem-solving. This project stands as a testament to our dedication to practical application and has undoubtedly enriched our understanding of robotics.